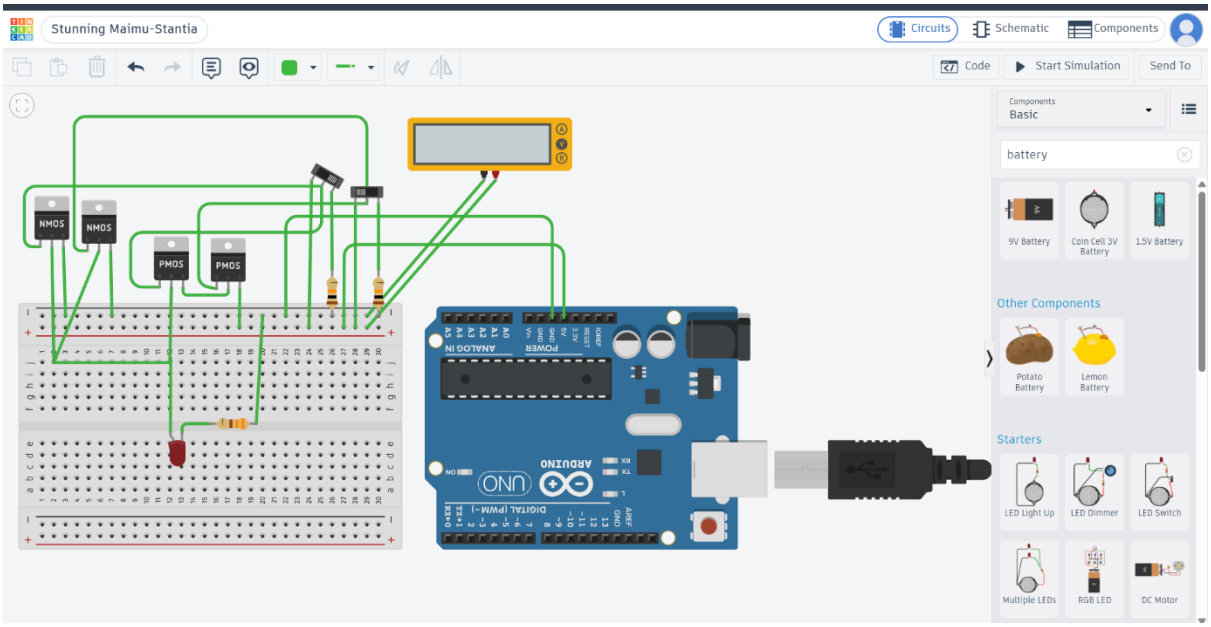
GROUP 37

PROJECT STATEMENT – Construct and simulate a NOR gate using MOS transistors, infer its behavior, and explain its truth table with logical operation.

Tinkercad Design



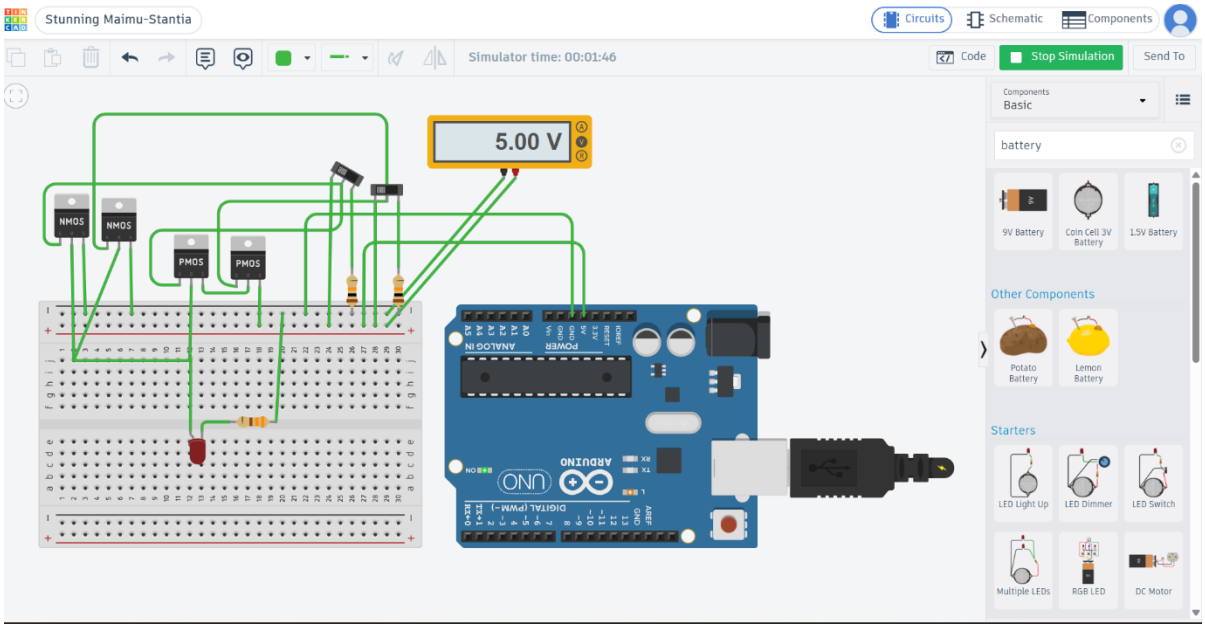
Inferring its behavior

The circuit exhibits **inverting dominance logic**. This means a single input signal acting as a **1 (HIGH / 5V)** completely dominates the circuit, forcing the output to **0 (LOW / 0V)**. The output is highly sensitive to any active high state:

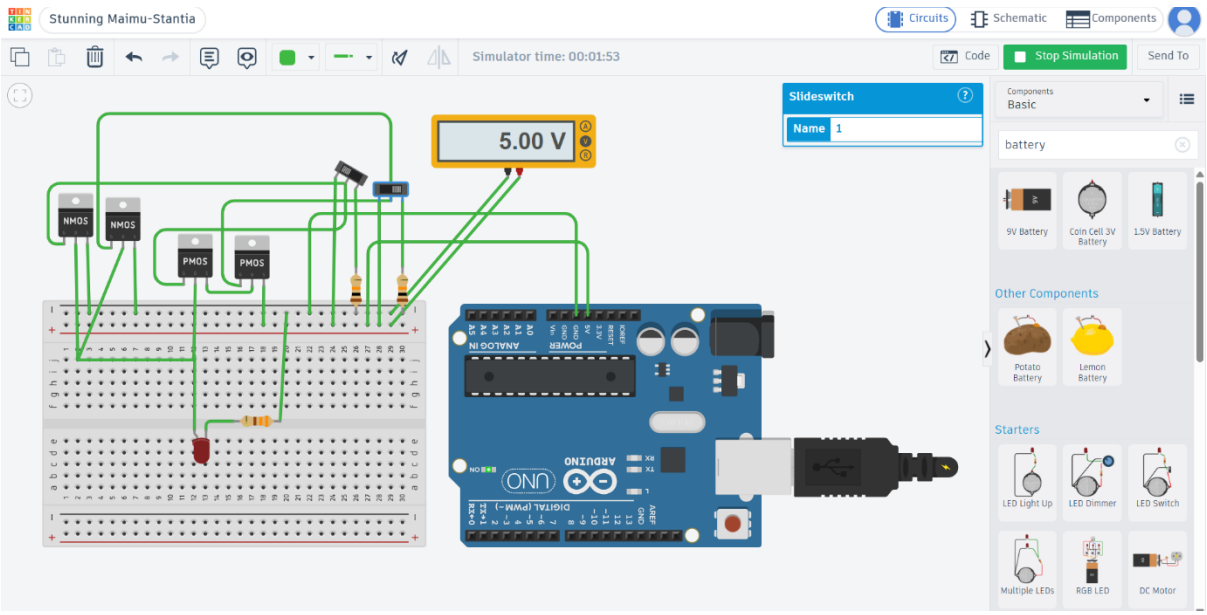
- **The Default State (Switches Right):** When both switches sit to the right, they route the 0V ground potential (via the 10kΩ pull-down resistors) to the transistor gates. In this configuration, the circuit behaves like a closed pipe connected to the 5V reservoir, letting current flood through the LED.
- **The Interrupted State (Switches Left):** Sliding either switch to the left floods that respective transistor gate path with 5V. The circuit responds instantaneously by severing the 5V supply line and opening a direct drainage pathway to Ground, instantly starving the LED of current.

Steps of Simulation

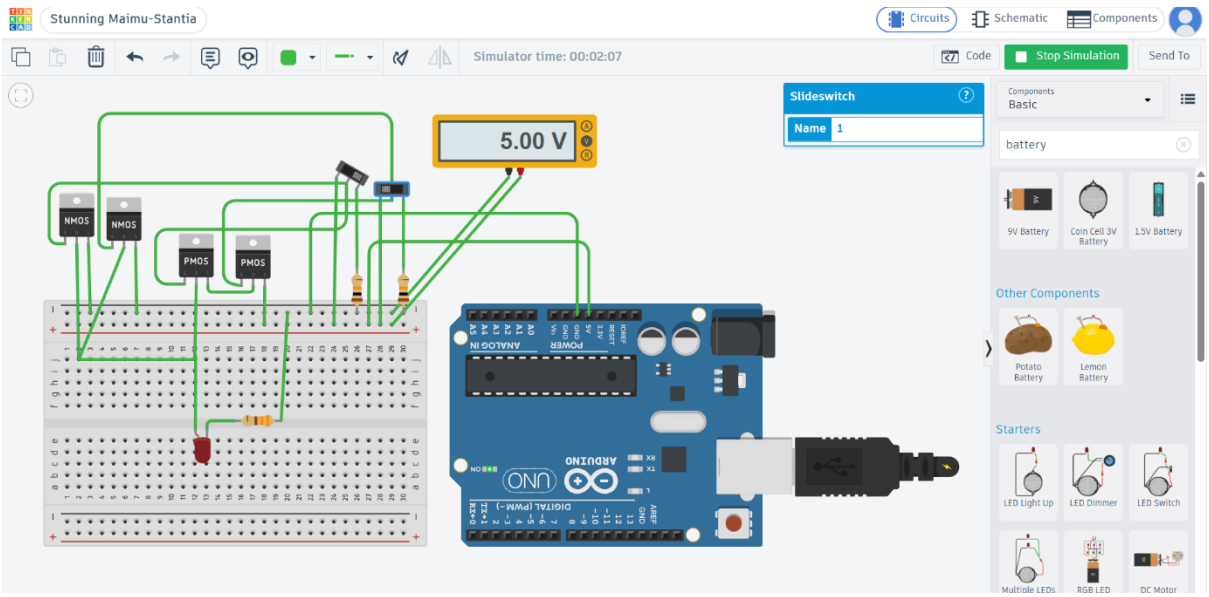
1.Both switches to the left (LED off)



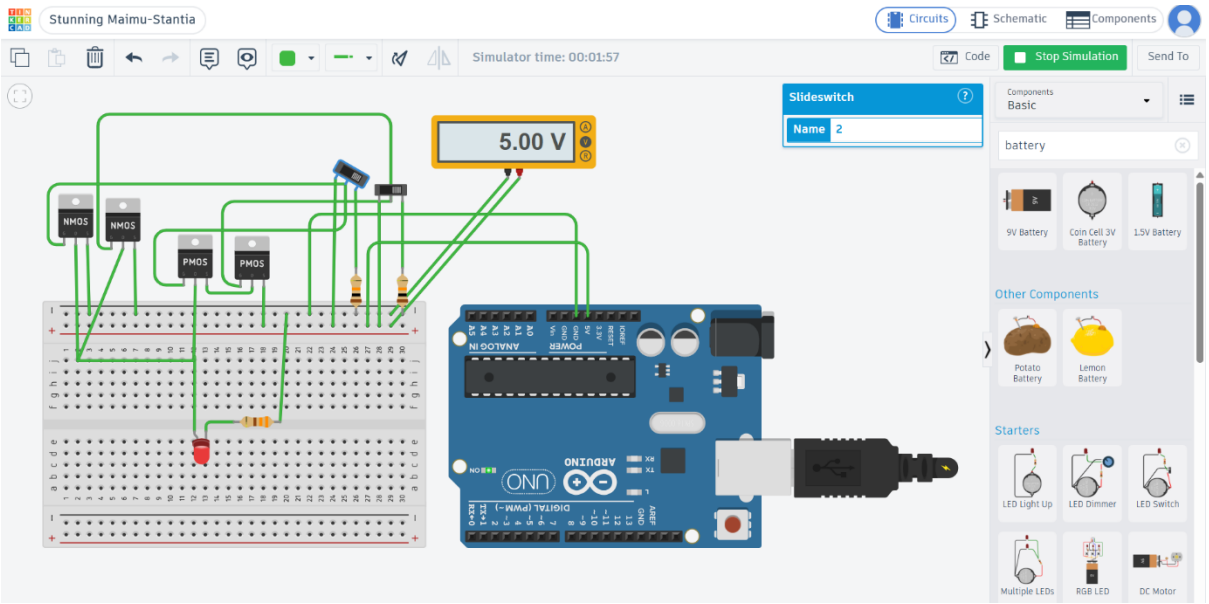
2.One switch to the left, other to the right (LED off)



3. One switch to the right, other to the left (LED off)



4. Both switches to the right (LED on)



Truth Table

Input A (Switch 1)	Input B (Switch 2)	PMOS A (Top)	PMOS B (Bottom)	NMOS A (Left)	NMOS B (Right)	Output Voltage	LED State
0 (Right)	**0** (Right)	**ON** (Closed)	**ON** (Closed)	OFF (Open)	OFF (Open)	**5.00 V**	**ON**
0 (Right)	**1** (Left)	**ON** (Closed)	OFF (Open)	OFF (Open)	**ON** (Closed)	**0.00 V**	**OFF**
1 (Left)	**0** (Right)	OFF (Open)	**ON** (Closed)	**ON** (Closed)	OFF (Open)	**0.00 V**	**OFF**
1 (Left)	**1** (Left)	OFF (Open)	OFF (Open)	**ON** (Closed)	**ON** (Closed)	**0.00 V**	**OFF**

Logic

The mathematical representation of the circuit is the NOR operation, represented algebraically as:

$$\text{Output} = A + B$$